

Elgamal and Schnorr scheme of Digital Signature

1. Elgamal Scheme (Digital Signature Scheme)

- This scheme is variant of **digital signature algorithm**.
- This scheme is based on computing assumption of large prime number.
- It is computationally very complex to compute S_1 and S_2 .
- This scheme assures that authenticity of message m sent by sender/signer to verifier.
- As with Elgamal encryption, the global elements of Elgamal digital signature are based on **prime number q and α , which is a primitive root of q** .

Algorithm

Generating private key & public key pair:

- **Step-1:** Generate a random integer X_A , such that $1 < X_A < q-1$.
- **Step-2:** Compute $Y_A = \alpha^{X_A} \bmod q$.
- **Step-3:** A 's private key is X_A ; A 's public key is $\{q, \alpha, Y_A\}$.

Create Digital Signature:

- **Step-1:** Choose a random integer K such that $1 \leq K \leq q-1$ and $\gcd(K, q-1) = 1$.
 K is relatively prime to $q-1$.
- **Step-2:** Compute $S_1 = \alpha^K \bmod q$.
- **Step-3:** Compute $S_2 = K^{-1}(m - X_A S_1) \bmod (q-1)$.
- **Step-4:** The signature consists of the pair (S_1, S_2) .

Signature Verification:

- **Step-1:** Calculate $V_1 = \alpha^m \bmod q$.
- **Step-2:** Calculate $V_2 = (Y_A)^{S_1} (S_1)^{S_2} \bmod q$.

2. Schnorr (Digital Signature Scheme):

- The Schnorr signature scheme is also based on discrete logarithms.
- The Schnorr scheme minimizes the message-dependent amount of computation required to generate a signature.
- The main work for signature generation does not depend on the message.
- The scheme is based on using a prime modulus p , with having a $(p-1)$ prime factor of q appropriate size; that is, $p \equiv 1 \pmod{q}$.
- Typically, we use $p = 2^{1024}$ and $q = 2^{160}$.
- Thus, p is a 1024-bit number, and q is a 160-bit number, which is also the length of the SHA-1 hash value.

Algorithm

Generating private key & public key pair:

Step-1: Choose primes p and q , such that q is a prime factor of $p-1$.

Step-2: Choose an integer α , such that $\alpha^q \equiv 1 \pmod{p}$.

(The values α , p , and q comprise a global public key that can be common to a group of users.)

Step-3: Choose a random integer s with $0 < s < q$.

This is the **user's private key**.

Step-4: Calculate $v = \alpha^{-s} \pmod{p}$.

This is the **user's public key**.

Create Digital Signature:

Step-1:

- Choose a random integer r with $0 < r < q$
- Compute $x = \alpha^r \pmod{p}$.
(This computation is a pre-processing stage independent of the message M to be signed.)

Step-2: Concatenate the message with and hash the result to compute the value:

$$e = H(M || x)$$

Step-3: Compute $y = (r + se) \bmod q$. The signature consists of the pair (e, y) .

Signature Verification

Step-1: Compute x'

$$x' = \alpha^y v^e \bmod p$$

$$x' = \alpha^y \alpha^{-se} \bmod p \quad (\because v = \alpha^{-s} \bmod p)$$

$$x' = \alpha^{(y-se)} \bmod p$$

$$x' = \alpha^r \bmod p \quad (\because y = r + se)$$

$$x' = x$$

So, here $x' = x$.

Step-2: Verify $e = H(M || x)$.

Hence, $H(M || x') = H(M || x)$.